

Scalability & Future Plan

Date	29 September 2026
Team ID	6ab2a427dbd693aadc5a7d4b
Project Name	Student Enrollment & Course Management System (SECMS)
Maximum Marks	2 Marks

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single Server Deployment - Currently runs on one Node.js server only	If server down, entire enrollment stops. Cannot handle 500+ students logging during admission peak time	High
2	Limited File Storage - Student documents stored in local MySQL / local folder	Storage will fill fast, risk of data loss, no backup. Slow retrieval during verification	Medium
3	No Advanced Analytics - Only basic PDF/Excel reports, no AI based insights	Admin cannot predict seat demand, course popularity, or identify dropout risk students	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Node.js server, supports	Move to AWS Elastic Beanstalk + Auto Scaling Group

		max 100-200 concurrent users	+ Load Balancer. Support 10,000+ students during admission season. Use Redis cache for seat availability
2	Data Storage	MySQL local, documents in local disk, no backup	Migrate to AWS RDS MySQL with read replicas + S3 for documents with lifecycle policy. Daily automated backup + CloudFront CDN for fast access
3	Performance	No caching, every request hits DB, slow during peak	Implement Redis for seat count caching, MySQL indexing on course_id, student_id. Add PM2 clustering and Nginx reverse proxy for 3x faster response
4	Security	Basic JWT auth, no rate limiting, no encryption for docs	Add OAuth2 + 2FA for admin, AWS WAF + Rate Limiter to prevent DDoS during enrollment, S3 SSE encryption, GDPR compliant audit logs

Future Roadmap:

Phase	Timeline	Feature	Description
Phase 1	Next 3 Months	Mobile App & Notifications	Launch React Native mobile app for students. Push notification via Firebase for enrollment deadline, payment reminder, approval status
Phase 2	6 Months	AI Course Recommendation	Use ML model to recommend courses based on student marks, interest, past enrollment trends. Predict seat demand and auto-suggest new sections
Phase 3	12 Months	Blockchain Certificates & Payment Gateway	Issue tamper-proof blockchain certificates for completed courses. Integrate multiple payment gateways - UPI, Netbanking, EMI for fees. Multi-college support SaaS model

